VINAY KUMAR DOKKU

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Summary

Software Developer with 6 years of experience designing and delivering scalable full-stack applications using Python, FastAPI, React.js, AWS, and Azure. Skilled in building secure APIs, architecting cloud-native backend services, and developing intuitive user interfaces that improve efficiency and user experience. Experienced in automating deployments with CI/CD pipelines, integrating AI-powered document processing, and ensuring high quality through unit and end-to-end testing. Known for driving impact by taking ownership of complex features, mentoring teammates, and collaborating across functions to deliver production-ready solutions in agile environments.

Skills

Programming Languages :	Python, Java, CoreJava, TypeScript, JavaScript, Node.js, React.js, Angular, HTML, CSS, PHP, SQL, MySQL, NoSQL, DynamoDB, MongoDB, Oracle, OOP.
Frameworks :	React.js, Angular 14, FastAPI, Flask, Django, Spring Boot, Chakra UI, SQLAlchemy.
Backend & APIs :	REST APIs, GraphQL (Ariadne), Middleware, Microservices, Event-driven Architecture.
Cloud Platforms :	AWS (ECS, ECR, DynamoDB, S3, IAM, SageMaker, Lambda, CloudWatch, CDK), Azure (Blob Storage, Key Vault, Cosmos DB, Functions), ETL/ELT Pipelines, Workflow Orchestration (Airflow, RDE Workflow).
DevOps & CI/CD :	Docker, Kubernetes, Terraform, GitLab CI/CD, Jenkins, Brazil-build, Hydra Tests.
Authentication-Security :	OIDC, SSO, AWS IAM, Azure Key Vault, RBAC, Token-based Authentication.
ML/AI & Automation :	OpenAI API, Azure OpenAI, Intelligent Document Processing, LLM Tagging, Selenium Automation, AI-powered Search, scikit-learn, Pandas, PySpark, ML Fundamentals (Regression, Clustering, ANOVA).
Testing & QA :	PyTest, Selenium WebDriver, Postman, Unit & Integration Testing, TDD.
Collaboration :	Agile/Scrum, Jira, Sprint Planning & Demos, Documentation, Knowledge Transfer, Mentorship.

Work Experience

Bank Of America, Software Developer — **February 2025 - Present**, Tampa, FL

PROJECT : Credit Risk – OneCredit Exposure Management

- Delivered microservices for global credit risk exposure tracking using Django, FastAPI, and Node.js and Angular,.
- Engineered scalable APIs to support IFRS, BASEL, CCAR/RWA, and SCCL regulatory data workflows.
- Implemented batch data pipelines using Apache Airflow and Node.js to process large-scale exposure and financial metrics.
- Optimized backend performance with SQLAlchemy and PostgreSQL, ensuring high-throughput, low-latency data operations.
- Leveraged and Dockerized services and configured CI/CD pipelines using Jenkins and Kubernetes to streamline deployments.
- Applied test-driven development with PyTest, Mocha, and Chai to ensure 90% test coverage across services.
- Integrated AWS services (Lambda, S3) for secure file storage and asynchronous processing tasks.
- Collaborated with product owners and business analysts to translate complex regulatory requirements into production-grade features.
- Contributed to Agile ceremonies, user story breakdowns, and sprint planning via Jira for effective team delivery.
- Spearheaded data driven microservice architecture enhancements, boosting regulatory compliance metrics delivery speed by 30% with new caching layer and API response optimizations using Redis and Nginx.

Boston Scientific, Software Developer — **August 2023 - Feb2025**, Boston, MA

PROJECT : PACE – Pricing and Contract Execution Platform

- Spearheaded development of an AI-powered contract intelligence platform, automating end-to-end lifecycle management for pricing, compliance, and reconciliation reducing manual effort and improving accuracy.
- Built scalable backend services with Python and FastAPI, and developed Angular 14 modules and reusable UI components with RxJS and Angular Material to deliver responsive, user-friendly dashboards.
- Designed and optimized data pipelines and models using PostgreSQL and DynamoDB, enabling seamless integration of structured and unstructured contract, invoice, and GPO data from Azure Blob Storage into CosmosDB.

- Implemented OpenAI-powered LLM workflows and a custom Impact Tracking engine to extract metadata from PDFs, enable automated compliance checks, and calculate revenue exposure from rule violations.
- Delivered GraphQL APIs and interactive dashboards for real-time visibility into contract status, violations, and GPO handling, enhancing decision-making for business stakeholders.
- Secured and optimized the platform through OIDC authentication, Azure Key Vault, automated testing with PyTest, and Terraform-based CI/CD pipelines, ensuring reliable, production-ready deployments.

PROJECT : Clinical AI – Intelligent PDF Extraction Platform

- Architected and delivered a full-stack AI solution to extract structured clinical data from unstructured PDF documents.
- Independently gathered requirements, proposed the tech stack (AWS EC2, SageMaker, Docker, React, Chakra UI), and led end-to-end implementation.
- Designed intuitive Figma mockups and converted them into a responsive React frontend with dynamic PDF upload and response rendering.
- Integrated OpenAI models via SageMaker to generate real-time metadata extraction based on uploaded clinical PDFs.
- Engineered a secure FastAPI backend to receive files, trigger AI inference, and return structured results.
- Containerized and deployed frontend and backend using Docker, hosted on AWS EC2 with IAM-secured access via Ubuntu and Putty.
- Built robust token validation, error handling, and response formatting to ensure stability and production readiness.
- Demonstrated the product through stakeholder demos and positioned it under the "Clinical AI" namespace for future expansion.

ScriptBees, Software Developer **January 2020 - June 2022**
Hyderabad, India

PROJECT : Sales Training Portal

- Architected the development of a Sales Training Portal, architecting a unified, AI-powered platform that integrated learning content from ALM, PodBean, and Vimeo into a centralized web and mobile-friendly solution.
- Designed and implemented the frontend using React.js and Chakra UI, delivering dynamic filtering, featured video tagging, and real-time search to improve content discovery across providers.
- Engineered REST API integrations with token-based authentication and automated ALM token refresh, ensuring secure, uninterrupted access and preparing the system for SSO enablement.
- Implemented caching strategies and optimized API call handling, reducing redundant requests by over 40% and improving portal responsiveness.
- Containerized applications with Docker, deployed on AWS EC2, and configured IAM roles and security policies to enforce access controls and compliance.
- Collaborated on Figma-driven UI designs, delivered live client demos, and incorporated feedback into iterative releases, leading to a smooth production rollout and increased adoption.

Education

Masters in Information Technology **GPA – 3.60**
Wilmington University · Newcastle, Delware · 2023

Ph.D. in Artificial Intelligence
University of the Cumberlands, Kentucky, USA · Present